

FAZLE RAFSANI

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An aspiring 2nd year computer science PhD student with medical imaging, deep learning, and computer vision experience. Experienced in creating generative and multimodal models for difficult imaging problems. Deep Learning enthusiast who aims to improve artificial intelligence for health care.

Education

Arizona State University

2nd year **Computer Science** PhD Student

Current CGPA **4.00 out of 4.00**

Fall 2023 – Ongoing

Tempe, AZ

Research Interest:

- Computer Vision
- Medical Imaging
- Machine Learning
- Deep Learning
- Generative Models
- Explainable AI

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Computer Science and Engineering

CGPA **3.54 out of 4.00**

February 2017 – May 2022

Dhaka

Technical Skills

Languages: Python, Java, C, C++, HTML/CSS, JavaScript, PHP, SQL

Python Libraries: Numpy, Pandas, Scikit-learn, Matplotlib, OpenCV, Pillow, Axios, Keras

Developer Tools: VS Code, Eclipse, Github, Git, Heroku, Docker, Kubernetes, SSH, Aws, Azure

ML Frameworks and tools: PyTorch, Keras, TensorFlow, HuggingFace, MONAI, GradCAM

Web Frameworks: Laravel, Django, Nodejs, React, Express, Nextjs

Experience

ASU-Mayo Center for Innovative Imaging Lab

Graduate Research Associate

January 2024 – present

Tempe, AZ

- Pursuing PhD in Computer Science at Arizona State University under the guidance of Dr. Teresa Wu and appointed as a Research Associate at the ASU-Mayo Center for Innovative Imaging Lab
- Leveraging **generative models** like diffusion models and GANs to enhance brain tumor segmentation.
- Worked with medical image processing tools such as **FreeSurfer, BET, MONAI**
- Applied **multi-modal models** like **CLIP** and Biomed-CLIP for integrating diverse data sources.

Arizona State University

Graduate Teaching Associate

August 2023 – December 2023

Tempe, Az

- Served as a Teaching Associate for CSE 110, an introductory computer science course
- Delivered lectures across four sections, engaging and managing a cohort of over 350 students.
- Designed and developed programming problems for assignments, reinforcing key concepts and problem-solving skills.

IQVIA

Software Engineer AI/ML

September 2022 – July 2023

Dhaka, Bangladesh

- Worked remotely, leveraged **Python and machine learning frameworks** to develop and implement solutions for data-driven insights.
- Contributed to the **"Next Best Actions"** initiative by analyzing pharmaceutical data and generating actionable **recommendations** for clients.

ICT CELL BUET

Software Developer

December 2021 – September 2022

Dhaka, Bangladesh

- Built responsive and interactive system using **Django, React.js** integrated with **PostgreSQL** ensuring seamless user experience.
- Successfully delivered a comprehensive solution to support secured university financial oversight through a scalable and user-friendly system.

Research Works

Anomaly Segmentation using Diffusion Models December 2023 – present

- Addressed limitations of existing ”**weakly-supervised**” methods that still require costly pixel-level labels for hyperparameter tuning during inference, enhancing the cost-effectiveness of our approach.
- **Automated** the selection of noise scale, segmentation threshold, and guidance strength without pixel-level labels using the **unguided forward process**.
- Aggregated **anomaly maps** from the guided forward process to amplify the signal strength of anomalous regions, improving segmentation accuracy by achieving **dice score of 80%**.

Headache Detection Using Brain MRI March 2024 – present

- Leveraging a specialized CLIP-based model with **contrastive learning-based** pre-training, further fine-tuned with a unique **Mayo Clinic** dataset for improved diagnostic accuracy.
- Applied **GradCAM** for **biomarker extraction**, enabling visualization of relevant areas in MRI images and enhancing interpretability for medical professionals.
- Achieved superior performance (**95% AUC**) compared to existing methods for headache detection.

Deep Learning Based Virtual Resource Management For VNF March 2021 – April 2022

- Proposed deep learning-based forecasting algorithms to predict the necessary **Virtual Network Functions** (VNFs) and virtual CPU resources for each VNF to complete network services deployed as Service Function Chains (SFC).
- Utilized **LSTM**, **bidirectional-LSTM**, and **CNN-based** algorithms to forecast resource needs from univariate and multivariate network traffic data.
- Developed a resource allocation algorithm, **DeepVRM**, based on the forecasting models, enabling efficient resource distribution to support **seamless SFC** completion.

White Matter Hyper-Intensity Segmentation from Brain MRI May 2024 – Present

- Focused on white matter **hyper-intensity segmentation** from brain MRI scans
- Utilized **nn-UNet for 3D data segmentation**, specifically targeting small lesion areas with improved accuracy.
- Achieved a **68% Dice score** in segmenting small lesions, demonstrating promising results for this challenging task.
- Currently exploring methods for **segmenting unannotated data** , aiming to extend segmentation capabilities where labeled data is limited.

Selected Publications

AnoFPDM: Anomaly Segmentation with Forward Process of Diffusion Models for Brain MRI

Yiming Che, **Fazle Rafsani**, Jay Shah, Md Mahfuzur Rahman Siddiquee, Teresa Wu

Accepted for presentation at WACV 2025 and pre-print available on [arXiv](#)

DeepVRM: Deep Learning Based Virtual Resource Management for Energy Efficiency

Zaman, Zakia, Sabidur Rahman, **Fazle Rafsani**, Ishraq R. Rahman, and Mahmuda Naznin

[DOI](#) | *Journal of Network and Systems Management*

Awards and Recognition

Gerald Farin Memorial Fellowship

[2024-2025]

Awarded for academic excellence and research potential.

Dean's List

[2022]

Recognized for outstanding academic performance in the final year of undergraduate studies.

Finalist, Blockchain Olympiad

[2021]

Achieved finalist position in the Blockchain Olympiad organized by ICT Division Bangladesh.

Board Merit Scholarship

[2014 & 2016]

Received for exemplary results in both Secondary and Higher Secondary School Certificate exams.